Data_Analyst — Step 3

Master Python for Data Analysis

Detailed Learning Notes • CareerCompass

1. Learning Objective

Use Python, NumPy and pandas to clean, transform, analyze and automate repeatable data workflows.

2. Core Concepts — Detailed Breakdown

Python essentials

Functions, loops, comprehensions, dictionaries, exceptions, file handling and modules are the base.

NumPy

Learn arrays, shapes, indexing, slicing, vectorization and basic numerical operations.

pandas

Master Series/DataFrame operations, filtering, grouping, merging, reshaping, missing data and time-series basics.

Cleaning and validation

Create explicit rules for missing values, types, duplicates and invalid records.

Automation

Turn repeated notebook work into reusable scripts or functions and save outputs with clear names and metadata.

3. Recommended Learning Workflow

- Complete Python exercises.
- Practice NumPy arrays.
- Work through pandas fundamentals.
- Perform a cleaning task.
- Automate a repeated report step.

5. Hands-on Project / Practice

Build a monthly sales-report script. It should read CSV files, standardize columns, validate required fields, clean missing/invalid values, calculate KPIs, create summary tables and save a final Excel/CSV report.

6. Common Mistakes & Pitfalls

- Using apply for everything when vectorized operations exist.
- Mutating data without keeping a raw copy.
- Ignoring data types.
- Writing one giant notebook cell.
- Hard-coding file paths and column names without configuration.

7. Completion Checklist

- I can load and inspect data.
- I can filter, group and merge DataFrames.
- I can handle missing values intentionally.
- I can automate a repeatable report.
- I can explain the difference between raw and transformed data.

8. Interview / Assessment Questions

- What is a DataFrame?
- What is vectorization?
- How do merge keys affect row counts?
- How should missing values be handled?
- Why keep raw data unchanged?

9. Recommended References

pandas User Guide: https://pandas.pydata.org/docs/user_guide/

Python for Data Science: https://www.youtube.com/watch?v=CMEWVn1uZpQ

Data Analysis with Python: https://www.youtube.com/watch?v=GPVsHOlRBBI

Python Tutorial: https://docs.python.org/3/tutorial/

Note: These notes are designed as a structured learning guide. Use the references for deeper, current documentation and hands-on practice.